Rounding and Ordering on a Number Line
Answer Key

Grade 2–3 Place Value Through 1,000

Quick Answers

1. 50
2. 80
3. 300
4. 400
5. —
6. 370
7. 273, 307, 370
8. 700
9. —
10. 2, 8, 470
11. 481, 418, 184, 148
12. 287, 296, 312, 300, 300, 300

Curriculum

Level: Grade 2–3 Place Value Through 1,000
1.NBT / 2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 18 tagged checks.

1. Round 48 to the nearest ten.

Step 1. Find the two tens 48 sits between. Counting by tens: $40, 50$. So the left end is 40 and the right end is 50.
Step 2. The middle mark is halfway between them, at 45.
Step 3. 48 is past the middle mark, so it is closer to the right end. $\boxed{50}$
Why the picture helps: the student can see that 48 is only two hops from 50 but eight hops from 40.

2. Round 83 to the nearest ten.

Step 1. The two tens around 83 are 80 and 90, so those are the ends of the line.
Step 2. The middle mark is 85.
Step 3. 83 has not reached the middle mark yet, so it stays with the left end. $\boxed{80}$
Common wrong answer: 90, from rounding "up" out of habit. Rounding always goes to the *nearer* end, and on this line the nearer end is behind.

3. Round 260 to the nearest hundred.

Step 1. Counting by hundreds, 260 sits between 200 and 300. Those are the ends.
Step 2. The middle mark is 250.
Step 3. 260 is past 250, so it is closer to the right end. $\boxed{300}$
Same picture, longer hops: each short tick is now worth ten instead of one, but the question is still "which end is closer?"

4. Round 418 to the nearest hundred.

Step 1. 418 sits between 400 and 500.
Step 2. The middle mark is 450.
Step 3. 418 is barely past the left end and nowhere near the middle, so it stays at the left end. $\boxed{400}$

Note: the ones digit 8 is a distraction here. When rounding to the nearest hundred, only the position between 400 and 500 matters.

5. Write $<$ or $>$ between 536 and 563.

Step 1. Both numbers have 5 hundreds, so the hundreds place does not decide it. Compare the next place along: tens.
Step 2. 536 has 3 tens; 563 has 6 tens. Three tens is less than six tens, so 536 sits to the left of 563 on the number line.
Step 3. The number on the left is the smaller one. $\boxed{536 < 563}$

Reading the symbol: the small point of the symbol aims at the smaller number.

6. Round 365 to the nearest ten.

Step 1. 365 sits between 360 and 370, so those are the ends.
Step 2. The middle mark is 365 — the number lands exactly on it.
Step 3. A tie needs a rule, and the agreed rule is to take the bigger end. $\boxed{370}$

Say the rule out loud: "exactly halfway rounds up." It is a convention, not something the picture decides — both ends really are five hops away.

7. Order 307, 370 and 273 from least to greatest.

Step 1. Compare hundreds first. 273 has 2 hundreds; the other two have 3 hundreds. So 273 is the smallest and goes first.
Step 2. 307 and 370 both have 3 hundreds, so compare tens: 0 tens against 7 tens. 307 comes before 370.
Step 3. Reading the number line from left to right: $\boxed{273,\ 307,\ 370}$

Why these numbers: they use the same digits in different places, so the only way through is to compare place by place.

8. Round 650 to the nearest hundred.

Step 1. 650 sits between 600 and 700.
Step 2. The middle mark is 650, so this is another exact tie.
Step 3. By the halfway rule, take the bigger end. $\boxed{700}$

Check the pattern: it matches problem 6 — exactly halfway always goes to the larger benchmark, whether the benchmarks are tens or hundreds.

9. Mia wrote $409 > 490$. Correct her.

Step 1. Comparing starts at the biggest place, not the last digit. Both numbers have 4 hundreds, so move to the tens.
Step 2. 409 has 0 tens; 490 has 9 tens. So 409 lands just past 400 on the line, while 490 sits far to the right, near 500.
Step 3. The number further left is smaller. $\boxed{409 < 490}$

Mia compared the ones digits first. Comparing should start with the hundreds, then the tens, and only then the ones — a single one is worth far less than a single ten.

10. 472 between two tens: (a) hops past 470, (b) hops short of 480, (c) round to the nearest ten.

Step 1 (a). From 470, count on: $471, 472$ — two hops past the lower ten. $\boxed{2}$

Step 2 (b). From 472 up to 480: $473, 474, 475, 476, 477, 478, 479, 480$ — eight hops short of the upper ten. $\boxed{8}$

Step 3 (c). Two hops is a shorter trip than eight hops, so 472 is closer to the lower ten. $\boxed{470}$

This is what rounding means: not a digit trick, but "which benchmark am I nearer to". The digit rule (look at the ones place) is just a shortcut for this count.

11. Order 418, 481, 184 and 148 from greatest to least.

Step 1. Sort by hundreds first: 418 and 481 have 4 hundreds; 184 and 148 have 1 hundred. So both 4-hundreds numbers come first.

Step 2. Within the 400s, compare tens: 1 ten against 8 tens, so 481 is greater than 418.

Step 3. Within the 100s, compare tens: 8 tens against 4 tens, so 184 is greater than 148.

Step 4. Greatest to least: $\boxed{481,\ 418,\ 184,\ 148}$

Why it is a challenge: every number uses the digits 1, 4 and 8, so only place value tells them apart.

12. Challenge: Room A 287 cans, Room B 312 cans, Room C 296 cans.

Step 1 (a). Compare hundreds: 287 and 296 have 2 hundreds, 312 has 3. So 312 is the greatest. Between the other two, compare tens: 8 tens against 9 tens, so 287 is less than 296.

Step 2 (a). Least to greatest: $\boxed{287,\ 296,\ 312}$

Step 3 (b). Round each to the nearest hundred. 287 and 296 are both past the middle mark 250 on the 200-to-300 line, and 312 has not reached the middle mark 350 on the 300-to-400 line, so all three counts land on the same hundred. $\boxed{300}$

Step 4 (c). *Sample answer (accept any equivalent explanation):* "All three counts are close enough to 300 that rounding hides the difference between them. Rounding tells you about how many, so you have to go back to the exact numbers to say who collected the most — Room B, with 312."

Grading note for part (c): the sentence should say that rounding throws away the detail that separates the rooms, or that equal rounded values can come from different exact values. Marking this part is a judgement call — it is the one item on the sheet that is not machine-checked.